

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 3A

Describing data: centre, spread and shape

Bellerbys
Preparation for University Education

Name _____ Class _____ Date _____

EXERCISE A1

How long is a typical call?

Slides 10-11

A random sample of 12 phone-call lengths, in minutes:

1, 7, 4, 1, 2, 4, 3, 48, 3, 5, 3, 6

1. Find the mean, median and mode.

ORDERED VALUES

Measure	Working	Answer (minutes)
Mean		
Median		
Mode		

2. Which measure best describes a typical call? Give one reason.

EXERCISE A2

Correct it or delete it?

Slides 12-13

Start from the original 12 calls each time. Original total = 87 minutes.

Treat (a) and (b) separately. Reorder the values after each change.

(a) Correct a recording error	(b) Exclude an ineligible call
The 48-minute call should be 5 minutes. Find the new mean and median.	The 48-minute call was outside the observation period. Delete it. Find the new mean and median.
Ordered values:	Ordered values:
Mean:	Mean:
Median:	Median:

CHOOSING AND COMPARING MEASURES

EXERCISE A3

Typical salary and total payroll

Slides 14-15

A café employs 14 staff. A few staff earn much more than the others.

Monthly salary summary	Amount
Mean	£1,475
Median	£1,280

1. Which figure better describes a typical employee's salary? Explain.

2. Calculate the café's total monthly payroll. Show your working.

3. Explain why multiplying the median by 14 may give the wrong total.

EXERCISE A4

Same mean, same range

Slides 18-19

Dataset	Waiting times (minutes)	Mean	Range
A	6, 10, 10, 10, 14	10	8
B	6, 6, 10, 14, 14	10	8

Are the observations equally spread out? Explain using their positions.

Compare how many observations lie at the centre and at the endpoints.

QUARTILES AND CENTRAL SPREAD

Quick check from slide 26

Position 2.75 lies between values 10 and 18. Find the value there.

EXERCISE A5

Weekly exercise hours

Slides 28-30

The gym records weekly exercise hours for a sample of 16 members.

5, 1, 9, 3, 14, 6, 0, 11, 4, 18, 3, 8, 2, 12, 1, 7

1. Sort the observations. Write one value under each position.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16

2. Find Q_1 , the median and Q_3 . Show positions and interpolation.

Use the $n + 1$ position method from class. Keep fractional positions.

Measure	Position	Working and value (hours)
Lower quartile, Q_1		
Median, Q_2		
Upper quartile, Q_3		

3. Calculate the interquartile range (IQR). Explain what it describes.

Five-number summary (slide 30)

Minimum	Q_1	Median	Q_3	Maximum

HOW A CHANGE AFFECTS THE SUMMARIES

EXERCISE A6 One value changes Slides 31-32

Use the original gym data from A5. Replace 18 hours with 30 hours. Keep the other 15 observations.

Original total = 104 hours. Number of observations = 16.

- 1. Predict which summaries change before calculating.
- 2. Calculate the summaries for the original and revised datasets.

Measure	Prediction: change / same	Original data	Replace 18 with 30
Mean			
Median			
Range			
IQR			

Working: show how you update the total.

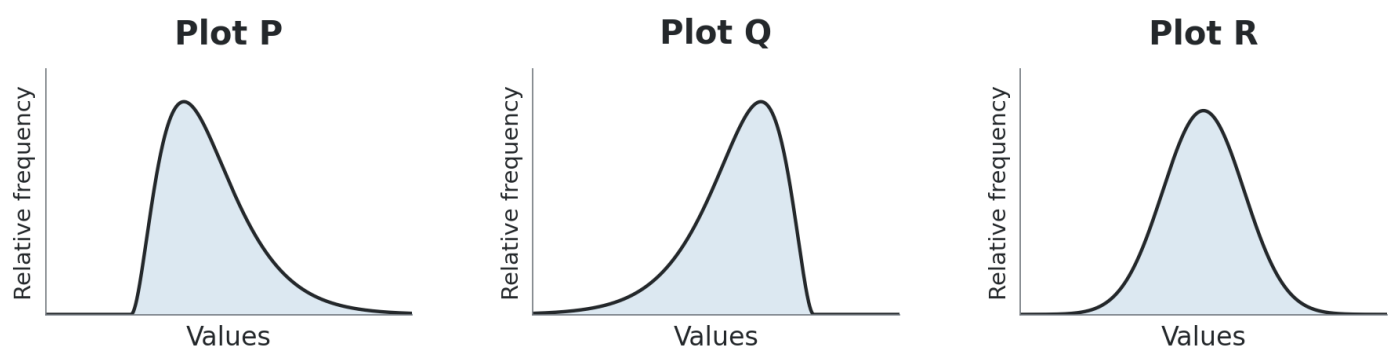
3. Which summaries stayed the same? Explain why.

STRETCH

Start again with the original dataset. Replace 18 hours with an unknown value so the new mean is 7.625 hours. Find the unknown value. Does the IQR change?

DISTRIBUTION SHAPE AND REASONING

EXERCISE A7 Shape detective Slides 36-39



1. Match each situation to P, Q or R. Name the shape and give a reason.

Situation	Plot and shape	Reason
Most calls are short. A few last much longer.		
Most test scores are high. A few are much lower.		
Package weights spread similarly either side of 500 g.		

2. A student looks at plot P and says:
“Most calls are short, so this is left-skewed.”

(a) Correct the explanation using the tail.

(b) For this shape, which is usually larger: mean or median? Why?

(c) Choose summaries for a typical call and the central spread. Explain.

INDIVIDUAL EXIT CHECK

EXERCISE A8 Calculate and explain Slides 41-42

Answer independently. Show your working and use the context.

1. There are 18 ordered observations. The 4th value is 51 and the 5th is 59.

Find Q_1 using our class method. Explain why “ $Q_1 = 4.75$ ” is incorrect.

Step	Your working
Quartile position	
Quartile value	

Explanation:

2. Two gyms have the same median. Their IQRs are 3 hours and 8.25 hours.

Which gym has less spread in the middle half of its observations? Explain.

3. A student says: “Deleting an extreme value can never change the median.”

Agree or disagree? Use A2 as evidence.